ALA: 2

Topic Name: Interview based questions

Sub-topic Name: Python syntax rules

Questions:

1. What is Python syntax?

- Python syntax refers to the set of rules that define how Python programs are written and interpreted by the Python interpreter.

Syntax:

```
variable_name = value
print(variable_name)
```

Example:

```
name = "Rahul"
age = 20
print("Name:", name)
print("Age:", age)
```

2. What is the purpose of the colon (:) in Python syntax?

- A colon (:) is used at the end of statements like if, for, while, function, and class definitions to start a block of code.

Example:

```
if x > 10:
    print("Greater")
```

3. What happens if indentation is incorrect in Python?

- If indentation is incorrect, Python raises an IndentationError, and the program will not run.

Syntax:

```
if condition:
    statement
```

Example:

```
x = 10
if x > 5:
print("x is greater than 5")
```

4. Why is indentation important in Python?

- In Python, indentation is used to define blocks of code (like loops, functions, and conditions). Unlike other languages that use { }, Python uses spaces or tabs to structure code.

Syntax:

(using if statement)

```
if condition:
    statement
```

Example:

(using if statement)

```
number = 10
if number > 5:
    print("Number is greater than 5")
```

5. What are the common Python syntax rules?

- Indentation, Case Sensitivity, Variable Naming Rules, Comments,Colon Usage, Statements, Multiple Statements in One Line, Keywords.

➢ **Indentation**

- **Syntax**
- if condition:
  statement

- **Example**
- x = 10

  if x > 5:
  print("x is greater than 5")

- **Output**
- x is greater than 5

➢ **Case Sensitivity**

- **Syntax**
- variable_name = value
  Variable_name = value

- **Example**
- name = "Shravya"
  Name = "Rahul"

  print(name)
  print(Name)

- **Output**
- Shravya
  Rahul

➢ **Variable Naming Rules**
- **Syntax**
- variable_name = value

- **Example**
- age = 20
  student_name = "Riya"

  print(age)
  print(student_name)

➢ **Comments**
- **Syntax**
- # comment text

- **Example**
- # This is a comment
  print("Hello Python")

- **Output**
- Hello Python

➢ **Colon Usage (:)**
- **Syntax**
- if condition:
  statement

- **Example**
- number = 8

  if number > 5:
  print("Number is greater than 5")

➢ **Statements**
- **Syntax**
- Statement

- **Example**
- x = 5
  y = 10
  print(x + y)

- **Output**

- 15

➢ **Multiple Statements in One Line**

- **Syntax**
- statement1; statement2; statement3
- **Example**

- a = 5; b = 10; print(a + b)

- **Output**
- 15

➢ **Keywords**

- **Syntax**
- if condition:
  statement

- **Example**
- x = 10

```
if x > 5:
    print("Greater")
else:
    print("Smaller")
```

7. What is the syntax for printing output in Python?

- **Syntax**

print(value)

Example

print("Hello Python")

Output

Hello Python

8. **What is the syntax rule for starting a code block using a colon?**

- **Syntax**

```
statement:
    block_of_code
```

Example

```
age = 20

if age > 18:
    print("Adult")
```

9. **What is the syntax rule for writing statements in Python?**

- **Syntax**

Statement

Example

```
x = 5
print(x)
```

10. **What is the syntax rule for writing multi-line comments in Python?**

- **Syntax**

```
"""
comment
"""
```

Example

```
"""
This is a
multi-line comment
"""
print("Python Syntax")
```